

BBE Sales Analytics

Business Analysis Report

Power BI Sales Performance Dashboard • Executive, Product, Customer, Geography & Promotion Analysis

Total Sales	\$22.20M	Gross Profit	\$9.20M
Gross Margin	41.46%	Total Orders	25,410
Customers	18,220	Average Order Value	\$873.76
Sales Lines	58,154	Analysis Period	Jan 2018 – Jan 2020

Purpose: provide a concise management-level interpretation of the cleaned sales dataset and the Power BI dashboard, with evidence-based observations and practical implications.

1. Executive Summary

The analysis covers sales activity from January 2018 through January 2020 and contains 58,154 cleaned sales-line records. The business generated approximately \$22.20M in sales, \$9.20M in gross profit and a 41.46% gross margin. Revenue is highly concentrated in the Bikes category, while North America is the largest territory group. A relatively small set of high-performing bike models accounts for a substantial share of revenue.

Management takeaway: protect and optimize the core bike portfolio while exploring opportunities to diversify accessories and clothing revenue and to reduce dependence on a small number of high-performing models and markets.

2. KPI Overview

Metric	Result	Interpretation
Total Sales	\$22.20M	Overall revenue recorded in the cleaned sales data
Gross Profit	\$9.20M	Sales less recorded ProductStandardCost
Gross Margin	41.46%	Gross profit as a percentage of sales
Total Orders	25,410	Distinct sales orders
Customers	18,220	Distinct customers appearing in sales
Average Order Value	\$873.76	Total sales divided by distinct orders

Metric	Result	Interpretation
Sales Lines	58,154	Transaction-line records after cleaning

3. Data Preparation & Quality

- **Duplicate handling:** 24 exact duplicate transaction rows were identified and removed, leaving 58,154 sales-line records.
- **Missing product keys:** 5 missing ProductKey values were recovered by matching the affected order-line records rather than deleting valid sales.
- **Relationship validation:** ProductKey, CustomerKey and PromotionKey relationships were validated against their respective dimension tables.
- **Data types:** dates, numeric fields and categorical fields were standardized for analysis.
- **Customer cleanup:** the Title field was removed because it was overwhelmingly null and did not add analytical value.
- **Optional attributes:** legitimate nulls in product/customer attributes were preserved rather than replaced with artificial values.

4. Data Model

The Power BI solution uses a star-schema approach. Fact_Sales contains the transaction grain, while Dim_Date, Dim_Product, Dim_Customer and Dim_Promotion provide descriptive dimensions. The primary date relationship uses OrderDate, with one-to-many relationships from each dimension to the sales fact.

Dimension	Key / Role	Fact relationship
Dim_Date	Date	Fact_Sales[OrderDate]
Dim_Product	ProductKey	Fact_Sales[ProductKey]
Dim_Customer	CustomerKey	Fact_Sales[CustomerKey]
Dim_Promotion	PromotionKey	Fact_Sales[PromotionKey]

5. Product & Category Analysis

Bikes are the dominant revenue category, generating approximately \$21.16M, or about 95.3% of total sales. Accessories contribute approximately \$0.70M and Clothing approximately \$0.34M. This concentration makes the bike portfolio the primary commercial engine of the dataset.

Category	Approx. Sales	Share of Total
Bikes	\$21.16M	95.3%
Accessories	\$0.70M	3.2%
Clothing	\$0.34M	1.5%

The leading subcategories are Road Bikes (~\$8.72M), Mountain Bikes (~\$8.60M) and Touring Bikes (~\$3.84M). Road and Mountain Bikes together represent roughly 78% of total sales. At model level, Mountain-200 and Road-250 are among the strongest revenue contributors.

Management implication: maintain inventory, pricing and availability discipline around the leading bike models, while using lower-revenue categories as potential areas for cross-selling and diversification.

6. Customer & Geographic Analysis

Geographic performance is measured using SalesTerritoryGroup and SalesTerritoryCountry in the sales fact table. North America is the largest territory group at approximately \$8.30M, followed by Europe at \$7.43M and Pacific at \$6.47M.

Territory Group	Approx. Sales
North America	\$8.30M
Europe	\$7.43M
Pacific	\$6.47M

At country level, the United States is the largest individual market at approximately \$6.91M, followed by Australia at \$6.47M and the United Kingdom at approximately \$2.84M.

Management implication: North America should remain a priority market, while Europe and Pacific provide meaningful revenue bases that can support geographic diversification.

7. Promotion Analysis

Most recorded sales occurred without a promotional discount, with approximately \$20.17M associated with the no-discount condition. The largest promotional sales association is the Volume Discount 11 to 14 promotion at approximately \$1.99M. Other named promotions account for much smaller sales amounts.

Interpretation caution: the dashboard identifies sales associated with promotions; it does not establish that a promotion caused a change in sales. A causal promotion study would require a controlled comparison, baseline period or additional experimental data.

8. Key Business Insights

- **Revenue concentration:** the business is overwhelmingly dependent on Bikes, creating a clear concentration risk but also a strong core franchise.
- **Product concentration:** Road and Mountain Bikes are the principal subcategory revenue drivers, with a small group of models producing substantial sales.
- **Geographic balance:** all three territory groups contribute materially, although North America is the largest.
- **Customer scale:** 18,220 customers and 25,410 orders indicate a broad customer base relative to the overall sales volume.
- **Promotion mix:** the majority of sales are associated with no discount, suggesting that baseline demand is not dependent on recorded promotions in this dataset.
- **Profitability:** the overall 41.46% gross margin provides a useful benchmark for comparing products, categories and promotions.

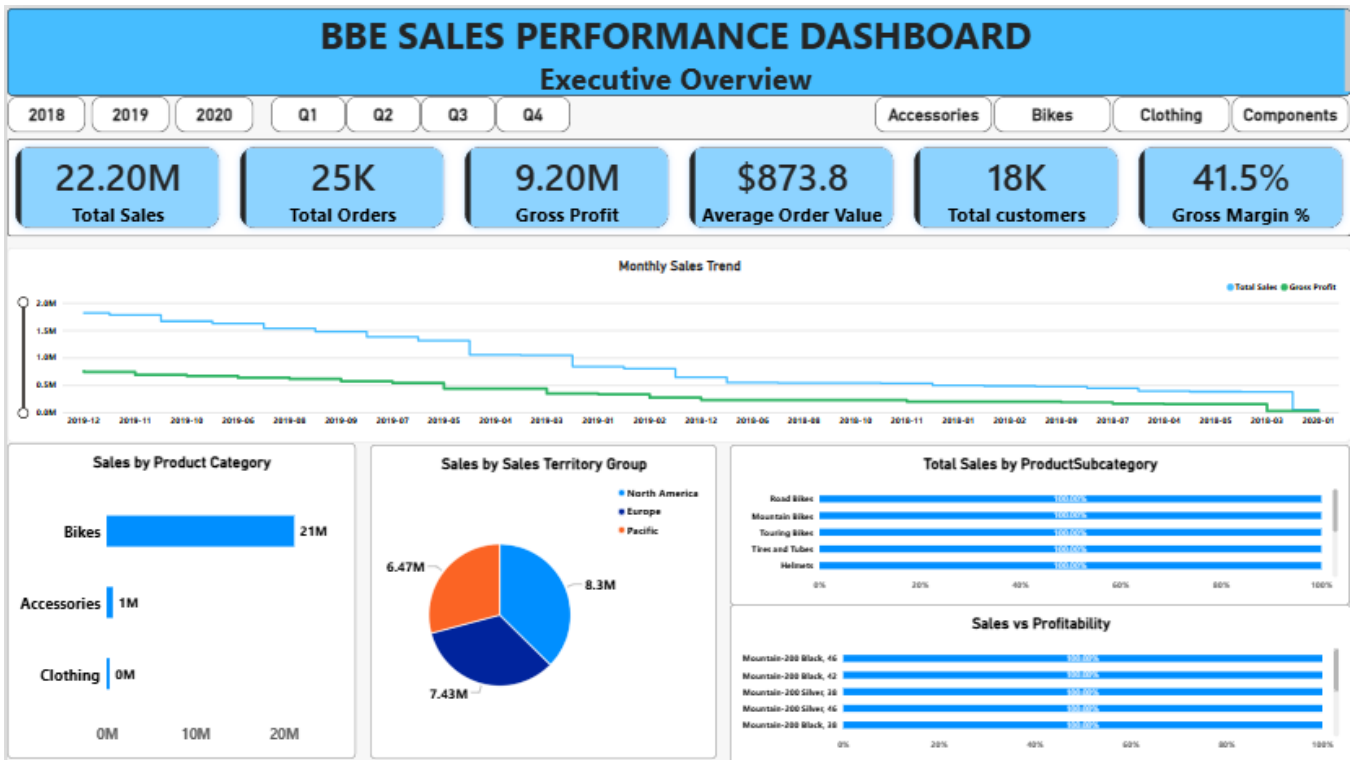
9. Management Recommendations

- **Protect the core:** prioritize availability, pricing and inventory planning for the highest-revenue bike models.
- **Diversify revenue:** investigate cross-selling strategies for Accessories and Clothing to reduce category concentration.

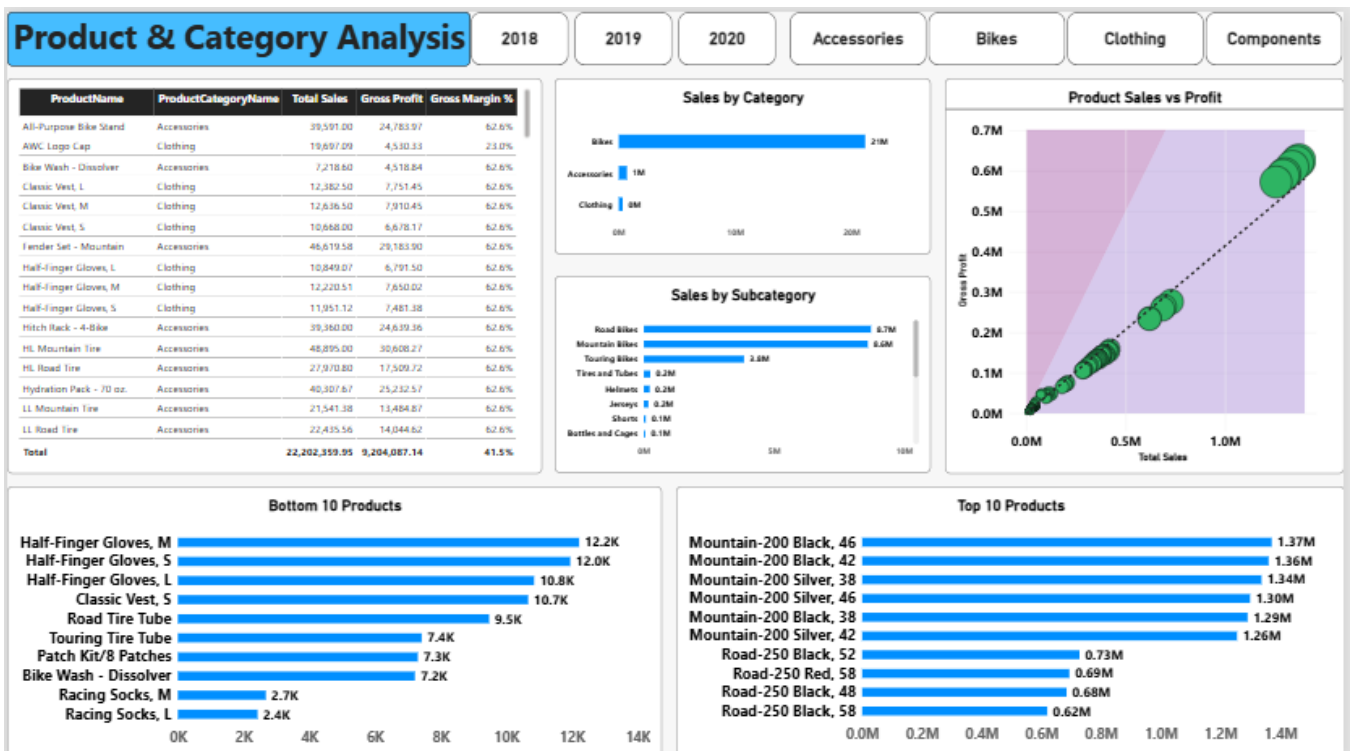
- **Use margin as a decision lens:** evaluate products and promotions using both sales and gross margin rather than revenue alone.
- **Prioritize key markets:** maintain strong execution in North America while identifying growth opportunities in Europe and Pacific.
- **Evaluate promotions rigorously:** compare promoted and non-promoted periods/products before increasing discount depth or frequency.
- **Monitor concentration:** track the share of sales generated by the top products, subcategories and countries as a recurring management KPI.

10. Dashboard Screenshots

Executive Overview



Product & Category Analysis



11. Conclusion

The BBE Sales Analytics solution converts the raw transaction dataset into a structured Power BI reporting model that supports executive monitoring, product analysis, customer/geographic analysis and promotion review. The strongest commercial signal is the dominance of the Bikes category and a concentrated set of high-performing models, combined with meaningful contributions from North America, Europe and Pacific.

The recommended management focus is therefore twofold: protect the high-performing core through disciplined product and market management, while developing diversification opportunities in lower-revenue categories and testing promotions using evidence-based performance comparisons.

Prepared for client delivery — BBE Sales Analytics Dashboard